APPLIED SCIENCE & THERMODYNAMICS FOR ENGINEERING (à à â¶flà â â¶"à à)

Thermodynamics, Heat Transfer, Electromagnetism, Refrigerant Chemistry & Pressure Laws

1. LAWS OF THERMODYNAMICS & HEAT TRANSFER

PHYSICS OF HEAT MOVEMENT & PHASE TRANSFORMATION:
* 1st Law of Thermodynamics (Conservation): Energy cannot be created or destroyed, only transformed.
* 2nd Law of Thermodynamics: Heat flows spontaneously from a warmer body to a colder body.
* In a refrigerator, external work (compressor electric power) is required to pump heat backwards.
* Conduction: Heat transfer through direct molecular collision (copper tube walls, heat exchangers).
* Convection: Heat transfer via moving fluids or gases (evaporator blower air, liquid refrigerant).
* Radiation: Heat transfer via electromagnetic infrared waves (sunlight through window glass).

2. PRESSURE - TEMPERATURE SATURATION RELATIONSHIP

BOILING POINT VARIATION WITH APPLIED PRESSURE:
* Pure water boils at 100 deg C at 1 atm (14.7 PSIA / 0 PSIG). At high vacuum (500 microns), water boils at -20 deg C!
* This is why deep vacuum pumps are used to vaporize and extract liquid moisture from AC pipework.
* Superheat = Actual Suction Pipe Temp - Saturation Suction Temp (Protects compressor from liquid slugging).
* Subcooling = Condenser Saturation Temp - Liquid Line Temp (Ensures 100% solid liquid reaches metering device).

3. REFRIGERANT CHEMISTRY & ENVIRONMENTAL IMPACT

MONTREAL & KIGALI PROTOCOLS - SAFETY CLASSIFICATION:
* ODP (Ozone Depletion Potential): Chlorine atoms destroy stratospheric ozone (CFCs & HCFCs such as R22).
* GWP (Global Warming Potential): Greenhouse effect rating relative to CO2 (R410A GWP = 2088, R32 GWP = 675).
* ASHRAE Safety Groups: A1 = Non-toxic, Non-flammable (R134a, R410A) | A2L = Mildly Flammable (R32, R1234yf)
* A3 = Highly Flammable Hydrocarbons (R600a Isobutane, R290 Propane) - Requires EX-rated safety sparkless tools.